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Chapter 9

The Capital Asset Pricing Model

INVESTMENTS
THIRTEENTH EDITION
BODIE, KANE, MARCUS

Capital Asset Pricing Model (CAPM)

- The **CAPM** is a set of predictions concerning equilibrium expected returns on risky assets.
- Based on two sets of assumptions.
  - Individual behavior.
  - Market structure.
- Markowitz established modern portfolio management in 1952.
- Sharpe, Lintner and Mossin published CAPM in 1964.
- 1990 Nobel Prize for economics: Harry Markowitz, Merton Miller and William Sharpe

Table 9.1 Assumptions

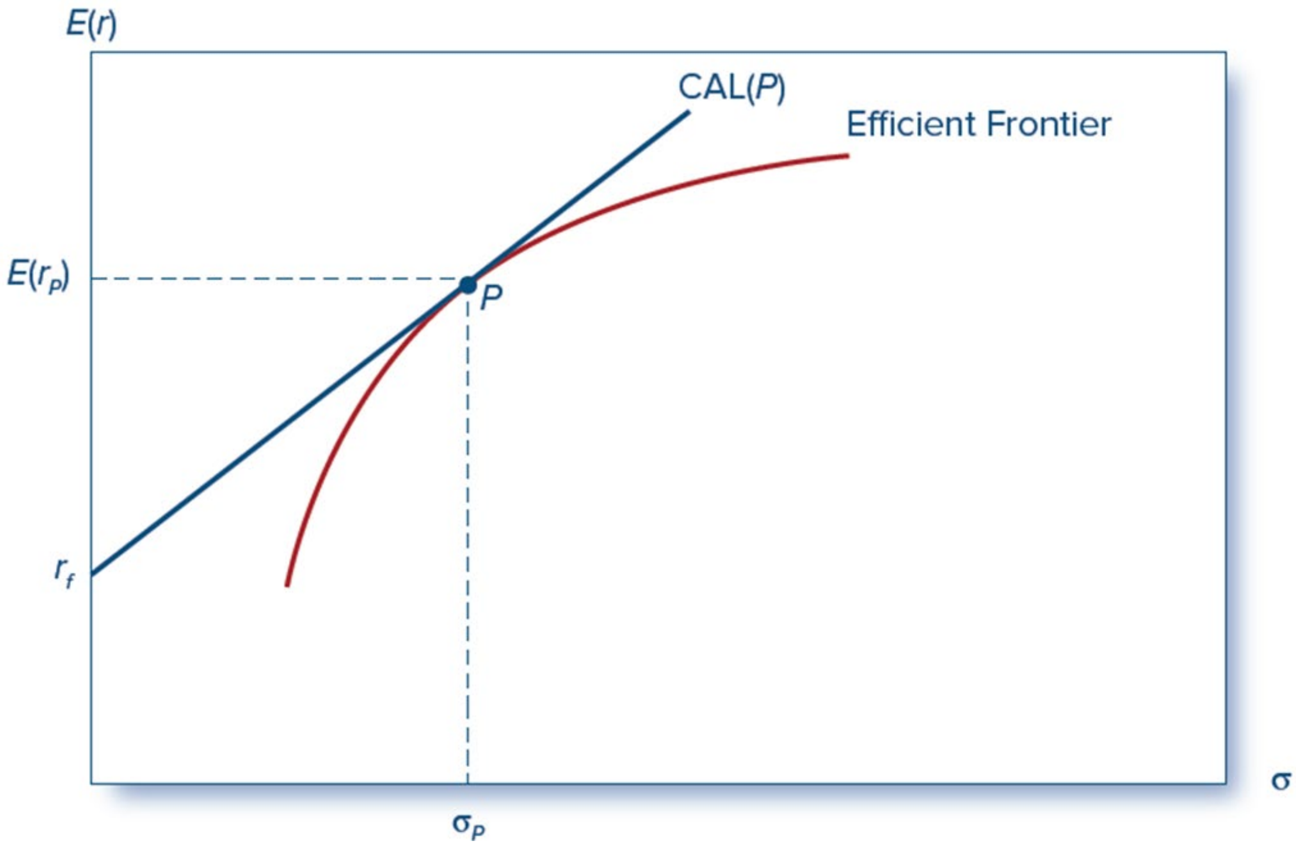
1. Individual behavior.
   a. Investors are rational, mean–variance optimizers.
   b. Their common planning horizon is a single period.
   c. Investors all use identical input lists, an assumption often termed **homogeneous expectations**. Homogeneous expectations are consistent with the assumption that all-relevant information is publicly available.
2. Market structure.
   a. All assets are publicly held and trade on public exchanges.
   b. Investors can borrow or lend at a common risk-free rate, and they can take short positions on traded securities.
   c. No taxes.
   d. No trading costs.

The Market Portfolio

- All investors will hold the same portfolio for risky assets—**market portfolio.**
- Market portfolio contains all securities
  - Proportion of each stock in this portfolio equals the market value of the stock ($P_{\text{Per Share}} \times Q_{\text{SharesOutstanding}}$) divided by the sum of the market value of all stocks.

Figure 9.1A Capital Allocation Line

A: The Efficient Frontier of Risky With the Optimal CAL.

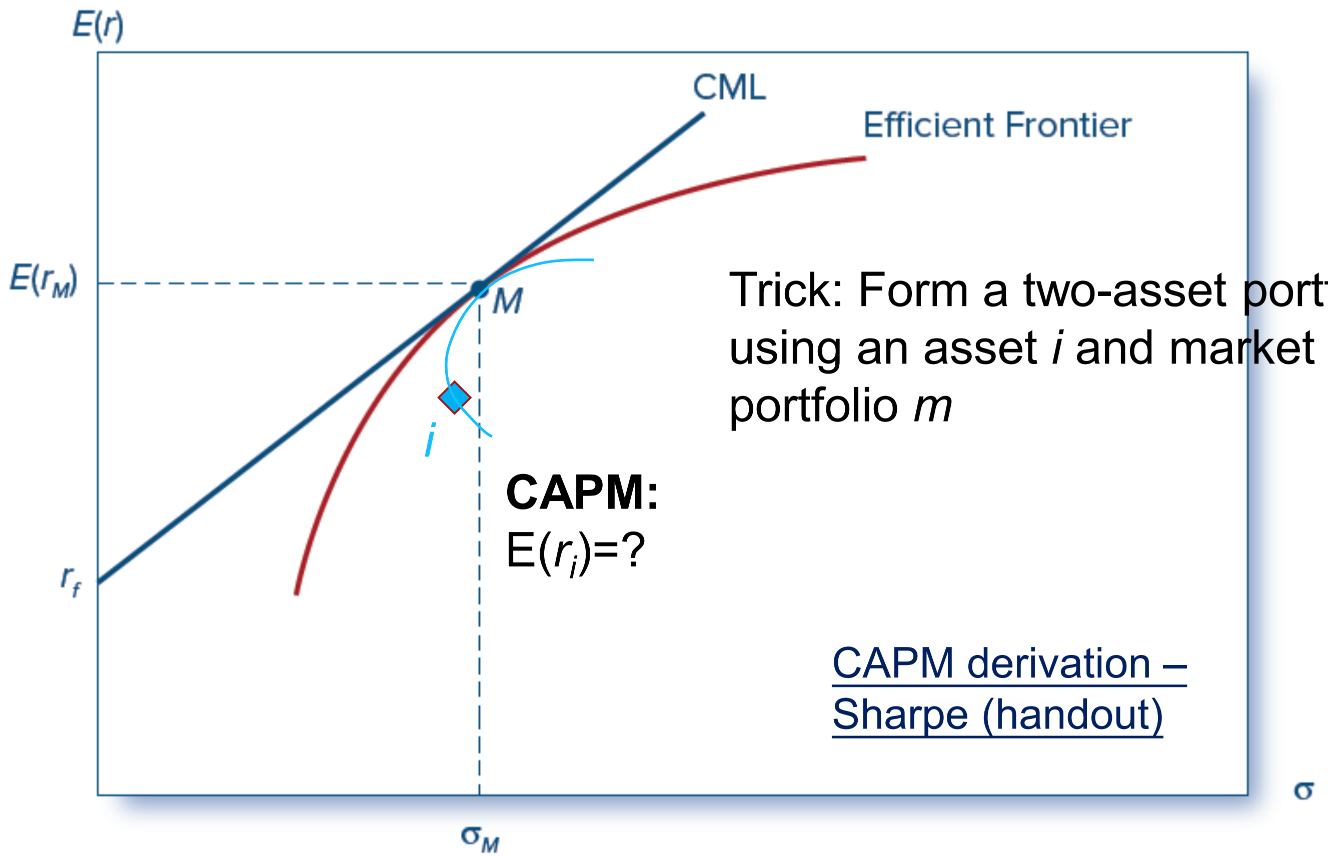


Why All Investors Would Hold the Market Portfolio

- Given all assumptions, all investors derive identical efficient frontiers and find the same tangent portfolio for the CAL.
- Market is the aggregate of all individual portfolios.
- The price adjustment process guarantees that all stocks will be included in the optimal portfolio. The only issue is price.
- If all investors hold an identical risky portfolio, this portfolio must be the market portfolio.

Figure 9.1B Capital Market Line

B: The Efficient Frontier and the Capital Market Line



Trick: Form a two-asset portfolio using an asset *i* and market portfolio *m*

CAPM:
$E(r_i)$=?

[CAPM derivation – Sharpe (handout)]

The Risk Premium of the Market Portfolio 1

- Recall that each individual investor chooses a proportion y, allocated to the optimal portfolio M, such that:

$$y = \frac{E(r_M) - r_f}{A\sigma_M^2}$$

- Recall the market risk premium (expected excess return):

$$E(R_M) = E(r_M) - r_f$$

Where:

$E(r_M)$ = Expected return of the market

r_f = Risk-free return.

A = Coefficient of risk a version.

σ_M^2 = Variance of the market portfolio

The Risk Premium of the Market Portfolio 2

- Net borrowing and lending across all investors must be zero, therefore the average position in the risky portfolio is:

$$\bar{y} = 1$$

- Thus, the market risk premium is proportional to its risk and the representative investor's degree of risk aversion:

$$E(R_M) = \bar{A}\sigma_M^2$$

Where:

$\bar{A}$ = Representative investor's risk a version

σ_M^2 = Variance of the market portfolio

Expected Returns on Individual Securities

- CAPM is built on the insight: The appropriate risk premium on an asset will be determined by its *contribution to the risk* of investors' overall portfolios.
- All investors use the same input list

  → market is the optimal risky portfolio.

Individual Securities: GE Example

- The variance of the portfolio is:

The sum over all the elements of the covariance matrix multiplying the weight from the row and the column

Portfolio Weights	w_1	w_2	...	w_{GE}	...	w_n
w_1	$Cov(r_1, r_1)$	$Cov(r_1, r_2)$	...	$Cov(r_1, r_{GE})$	...	$Cov(r_1, r_n)$
w_2	$Cov(r_2, r_1)$	$Cov(r_2, r_2)$	...	$Cov(r_2, r_{GE})$	...	$Cov(r_2, r_n)$
⋮	⋮	⋮		⋮		⋮
w_{GE}	$Cov(r_{GE}, r_1)$	$Cov(r_{GE}, r_2)$	...	$Cov(r_{GE}, r_{GE})$	...	$Cov(r_{GE}, r_n)$
⋮	⋮	⋮		⋮		⋮
w_n	$Cov(r_n, r_1)$	$Cov(r_n, r_2)$	...	$Cov(r_n, r_{GE})$	...	$Cov(r_n, r_n)$

The **contribution** of GE's stock to the variance of the market portfolio

$$w_{GE}\sum_{i=1}^{n} w_i Cov(r_{GE}, r_i)$$

Also true for Excess return *R*

$$\boxed{w_{GE}\sum_{i=1}^{n} w_i Cov(R_{GE}, R_i)}$$

GE Example 1

- Covariance of GE return with the market portfolio:

$$\boxed{\sum_{i=1}^{n} w_i Cov\,(R_i\,, R_{GE})} = \sum_{i=1}^{n} Cov\,(w_i R_i\,, R_{GE}) = \boxed{Cov\left(\boxed{\sum_{i=1}^{n} w_i R_i}\,, R_{GE}\right)}$$

- reward-to-risk ratio for GE would be:

$$\frac{GE's\ contribution\ to\ risk\ premium}{GE's\ contribution\ to\ variance} = \frac{w_{GE} E(R_{GE})}{w_{GE} Cov(R_{GE}, R_M)} = \frac{E(R_{GE})}{\boxed{Cov(R_{GE}, \boxed{R_M})}}$$

GE Example 2

- Reward-to-risk ratio for investment in market portfolio (i.e., **market price of risk**):

$$\frac{\text{Market risk premium}}{\text{Market variance}} = \frac{E\left(R_M\right)}{\sigma_M^2}$$

- Equilibrium dictates all investments should offer the same reward-to-risk ratio:

$$\frac{E\left(R_{GE}\right)}{Cov\left(R_{Ge}, R_M\right)} = \frac{E\left(R_M\right)}{\sigma_M^2}$$

Note: Market Price of risk $\underbrace{\left(\frac{E\left(r_M\right) - r_f}{\sigma_M^2}\right) \neq \text{ Sharpe ratio } \left(\frac{E\left(r_M\right) - r_f}{\sigma_M}\right)}$

GE Example 3

- Fair risk premium for GE stock:

$$E\left(R_{GE}\right)=\frac{Cov\left(R_{Ge},R_{M}\right)}{\sigma^{2}\left(R_{M}\right)}E\left(R_{M}\right)$$

承擔非系統風險

No return

- Restating, we obtain:

$$E\left(r_{GE}\right)=r_{f}+\beta_{GE}\left[E\left(r_{M}\right)-r_{f}\right]$$

Expected Return–Beta Relationship

- **Expected return–beta relationship**
  - The total expected rate of return is the sum of the risk-free rate plus a risk premium.
  - Risk premium is the product of a “benchmark risk premium” and the relative risk of the particular asset as measured by its beta.

Expected Return-Beta Relationship

- CAPM holds for the overall portfolio because:

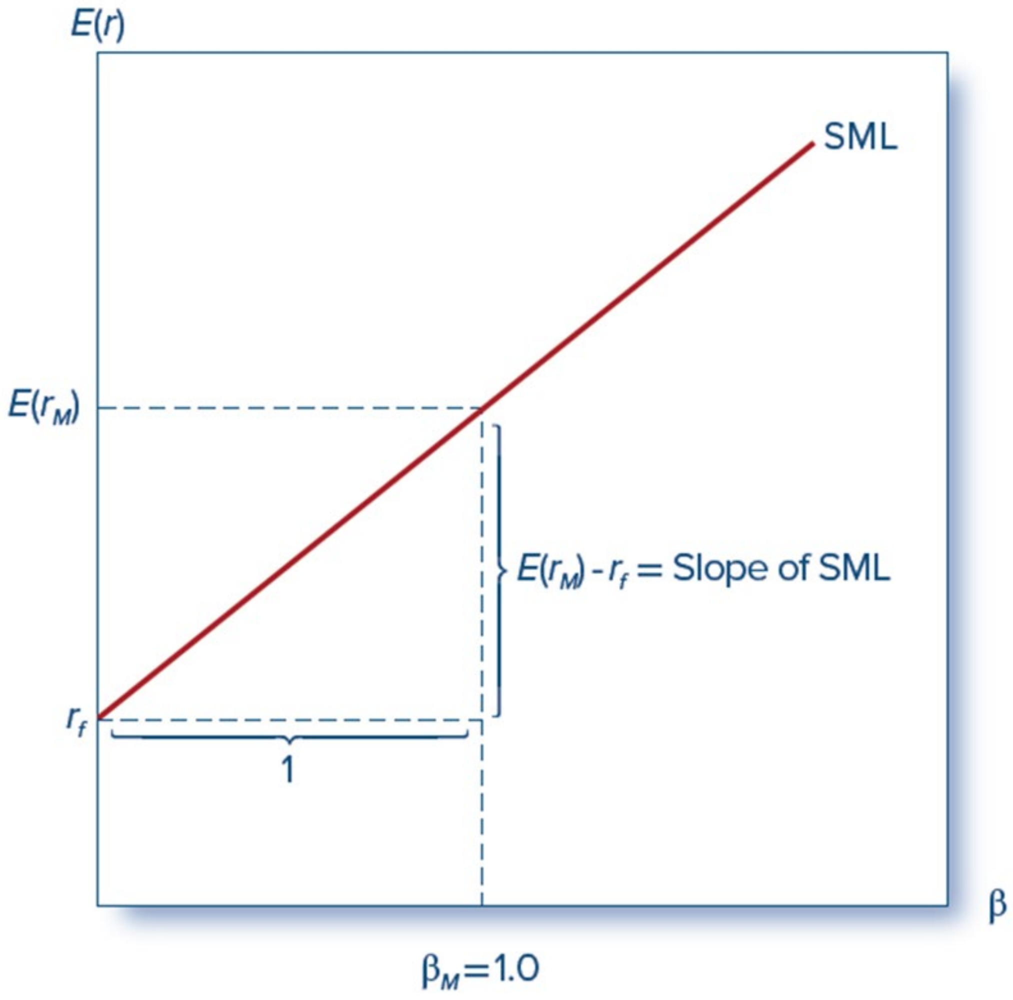
$$E(r_P) = \sum_k w_k E(r_k) \text{ and}$$

$$\beta_{\mathrm{P}} = \sum_k w_k \beta_k$$

- This also holds for the market portfolio:

$$E(r_M) = r_f + \beta_M \left[E(r_M) - r_f \right]$$

Figure 9.2 The Security Market Line



SML (Security Market Line) : The graphical representation of the expected return-beta relationship of the CAPM

The CAPM and the Single-Index Market

- Index model states the realized excess return on any stock is the sum of the following:
  - Realized excess return due to market-wide factors.
  - A nonmarket premium.
  - Firm-specific outcomes.

$$\underbrace{R_i}_{r_i - r_f} = \alpha_i + \beta_i \underbrace{R_M}_{r_m - r_f} + e_i$$

- The index model beta coefficient is the same as the beta of the CAPM expected return–beta relationship.

Alpha

異常報酬

- Alpha: The ***abnormal*** rate of return on a security in excess of what would be predicted by an equilibrium model such as CAPM (The *difference* between the *fair* return, $E(r_i)$ from CAPM, and *actually* expected rate of return on a stock)

$$R_i = \alpha_i + \beta_i R_M + e_i$$

Alpha and CAPM

$$R_{it} = \alpha_i + \beta_i R_{Mt} + e_{it}$$

$$r_{it} - r_{ft} = \alpha_i + \beta_i [r_{Mt} - r_{ft}] + e_{it}$$

Taking expectation

$$E(r_{it}) - r_{ft} = \alpha_i + \beta_i [E(r_{Mt}) - r_{ft}] + E[e_{it}]$$

$$E(r_{it}) = \alpha_i + r_{ft} + \beta_i [E(r_{Mt}) - r_{ft}] + 0$$

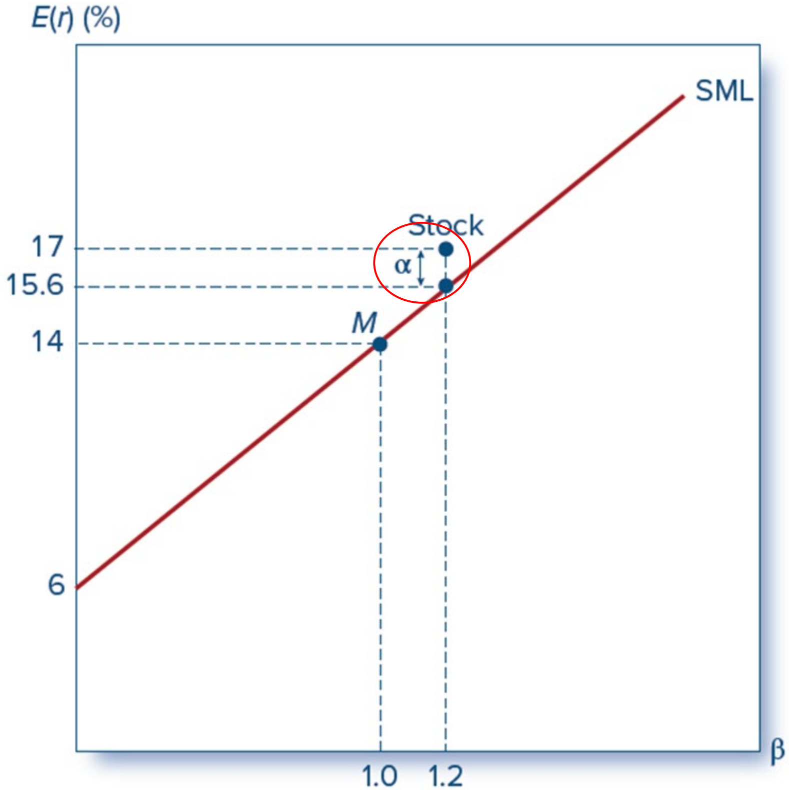
$$E(r_i) = \boxed{\alpha_i} + \boxed{\{r_f + \beta_i [E(r_M) - r_f]\}}$$

abnormal return

Expected return of CAPM

fair return

Figure 9.3 The SML and a Positive-Alpha Stock



Is the stock under or overpriced?

Underpriced : It is offering too high of a rate of return for its level of risk

能投機(定價太低)

The SML and a positive-alpha stock. The risk-free rate is 6%, and the market's expected return is 14%, implying a market risk premium of 8%.

Assumptions and Extensions of the CAPM

- Fundamental distinction between systematic and diversifiable risk remains in each variant of the basic model
- CAPM is built on “uncomfortably restrictive” assumptions

Recall Table 9.1:

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2.	Market structure.
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Extensions of the CAPM 1

1. Identical Input Lists: In the absence of private information, investors should assume alpha values are zero; however, reality limits this assumption of identical inputs:

Impediments to selling securities short exist (in turn limiting short selling's ability to prevent price run-ups).

- Unlimited short position liability.
- Limited supply of shares to be borrowed.
- Many investment companies outright prohibited form short sales.

Different investors realize different after-tax returns from the same stock.

Extensions of the CAPM 2

2. Risk-Free Borrowing and the Zero-Beta Model.

Higher rates on borrowing than lending $\rightarrow$ borrowers and landers arrive at different tangency portfolios.

- Any combination of tangency portfolios is on the efficient frontier.
- The market portfolio is an aggregation of efficient portfolios.
- Every tangency portfolio has a companion zero-beta portfolio on the bottom half of the frontier.

Risk-tolerant investors who otherwise wish leverage the tangency portfolio instead tilt toward high-beta stock $\rightarrow$ their prices will rise and risk premiums will fall.

Helps to explain positive alphas on low beta stocks and negative alphas on high beta stocks and a flatter SML.

Extensions of the CAPM 3

3. Labor Income and Other Nontraded Assets.

   - Many assets are not tradable (e.g., private businesses, human capital, earning power of individuals, etc.)

4. Merton's Multiperiod Model and Hedge Portfolios.

   - Investors should be more concerned with the stream of consumption that wealth can buy for them.
   - Intertemporal CAPM (ICAPM).

Extensions of the CAPM 4

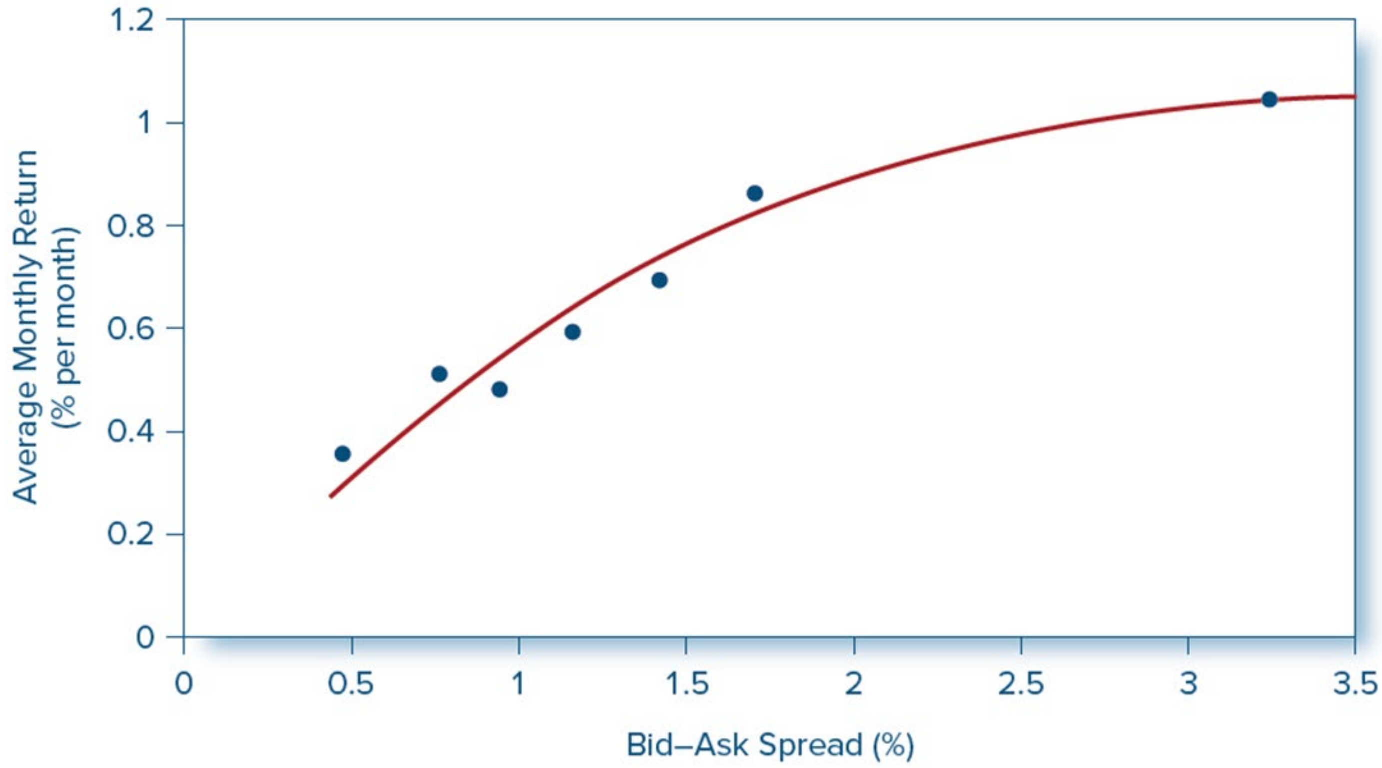
5. Consumption-Based CAPM (CCAPM).
   - Rubinstein, Lucas, and Breeden.
   - Investors allocate wealth between consumption today and investment for the future.

Extensions of the CAPM 5

6. Liquidity

- Financial costs inhibit trades.
- **Liquidity** of an asset is the ease and speed with which it can be sold at fair market value.
- **Illiquidity** can be measured in part by the discount from fair market value a seller must accept if the asset is to be sold quickly.

Figure 9.4 The Relationship Between Illiquidity and Average Returns



Source: Derived from Amihud, Y., & Mendelson, H. (1986). Asset pricing and the bid–ask spread. *Journal of Financial Economics, 17*, 223–249.

Liquidity Risk

- In a financial crisis, liquidity can unexpectedly dry up
- When liquidity in one stock decreases, it commonly tends to decrease in other stocks at the same time.
- Investors demand compensation for liquidity risk, demonstrated by firms with greater liquidity risk having higher average returns.
  - “Liquidity betas”

Issues in Testing the CAPM

- Testing the CAPM is surprisingly difficult.
  - Cannot observe all tradable assets.
    - Impossible to pin down market portfolio.
  - Both alpha and beta, as well as residual variance, are likely time varying.
- Most tests of the CAPM are directed at the mean–beta relationship as applied to assets with respect to an observed, but perhaps inefficient, stock index portfolio.

The CAPM and Investment Industry

- Portfolio theory and the CAPM have become accepted tools in the practitioner community.
  - Many professionals are comfortable with the use of beta to measure systematic risk.
  - Most investors don’t beat the index portfolio.